From Cells to Clusters

An Introduction to Single-Cell RNA-seq Analysis in Python

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github.com/Kur1sutaru/ISMB-LATAM

Today's plan



9:00 Introduction (slides)



9:20 00 · Cell Ranger & AnnData



9:40 01 · Quality control



10:15 02 · PCA, UMAP, clustering



10:45 Coffee break



11:00 03 · Cell type annotation



11:35 04 · Plots, DEGs, pathways



12:05 05 · Integration · R vs Python

~65 % hands-on in Google Colab · materials in English and Spanish

By 12:30 you will be able to…

- Go from a Cell Ranger matrix to an AnnData object
- Remove low-quality cells and doublets, normalize
- Reduce dimensions, build UMAPs, cluster cells
- Annotate cell types with markers, databases and atlases
- Make clear figures, find DEGs, interpret pathways
- Integrate samples, compare conditions, translate to R

Why single-cell? Smoothie vs. fruit salad

Bulk RNA-seq

- One average profile per sample
- A change could be a shift in expression or in cell composition, and you cannot tell which
- Rare cells (< 5 %) are invisible
- Cheap, deep, many replicates

Single-cell RNA-seq

- One profile per cell: thousands to millions of cells
- Discover cell types and states, including rare ones
- Measure composition and cell-type-specific responses
- Sparse (most values are 0), noisy, more expensive

Main technologies

Droplets

10x Chromium, Drop-seq. 1k–20k cells per channel, 3' or 5' counts with UMIs. The most common choice.

Plates

Smart-seq2/3. Hundreds of cells, full-length transcripts: isoforms, mutations, high sensitivity.

Combinatorial indexing

SPLiT-seq, Parse, sci-RNA-seq. Up to millions of cells and many samples, fixed cells.

Spatial

Visium HD, Xenium, MERFISH, CosMx. Expression with its position in the tissue.

Single-nucleus RNA-seq (snRNA-seq) works on frozen tissue and cells that are hard to dissociate (neurons, adipocytes); the analysis is almost identical.

How 10x Chromium works

1 · Encapsulate
Cell + barcoded gel bead in an oil droplet (GEM)

2 · Lyse & tag
mRNA captured by poly(dT); each molecule gets cell barcode + UMI

3 · Pool & amplify
Break droplets, reverse-transcribe, PCR, build the library

4 · Sequence
Illumina paired-end, ~20–50k reads per cell

Read structure (3' v3/v4)



Barcode → which cell

UMI → which molecule (removes PCR duplicates)

cDNA → which gene

Cell Ranger: from FASTQ to count matrix

Align
STAR to genome + annotation

Barcodes
Correct against the 10x whitelist

Count UMIs
Same barcode + UMI + gene = 1 molecule

Call cells
Separate cells from empty droplets

```
cellranger count --id=sample1 \
  --transcriptome=refdata-gex-GRCh38-2024-A \
  --fastqs=fastqs/ --sample=sample1 --create-bam=true
```

Needs a Linux server (≥ 64 GB RAM), not Colab. One run per sample; use cellranger multi for multiplexed or multi-modal libraries.



Open-source alternatives

- STARsolo
- alevin-fry / simpleaf
- kallisto | bustools
- Faster, free, run anywhere

What Cell Ranger gives you

File in outs/	Use it for
web_summary.html	QC report: always open it first
filtered_feature_bc_matrix(.h5)	Counts of called cells → our analysis
raw_feature_bc_matrix(.h5)	All barcodes → ambient RNA removal
metrics_summary.csv	Compare samples
possorted_genome_bam.bam	Velocity, genotyping (rarely)
cloupe.cloupe	Loupe Browser exploration

Check in web_summary.html

> 70 %

reads in cells

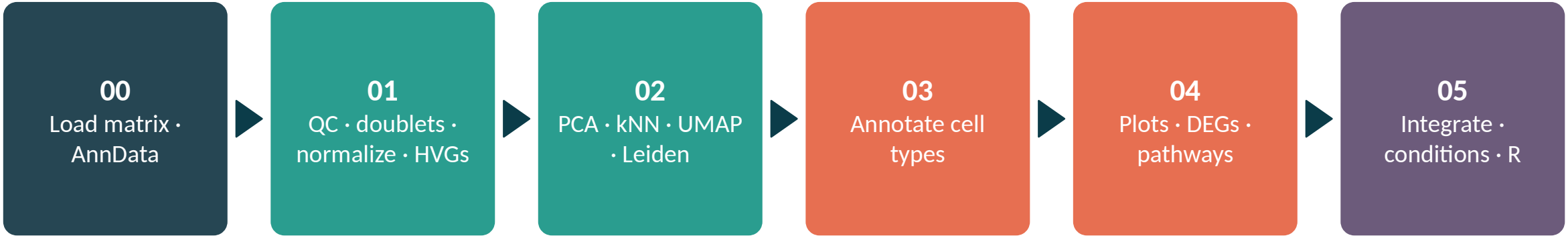
> 60 %

reads mapped to transcriptome

≥ 1,000

median genes per cell (tissue-dependent)

The workflow we follow today



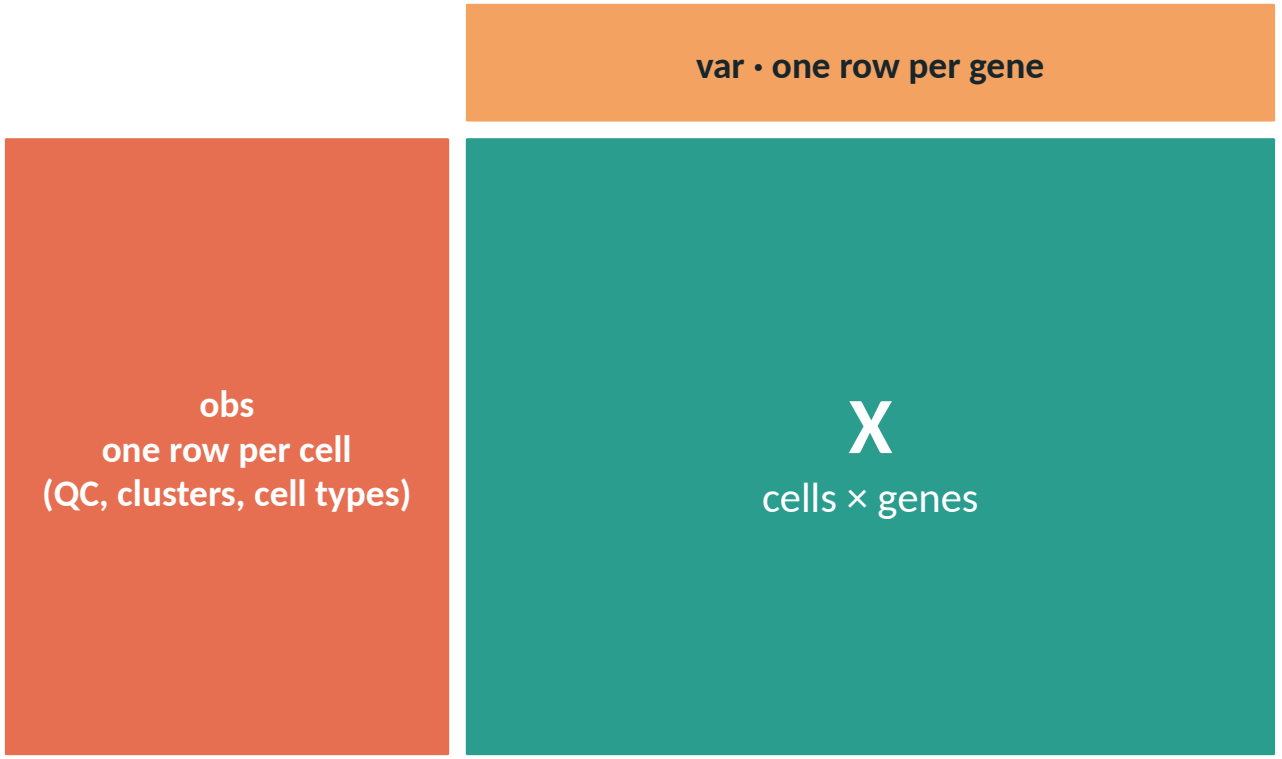
Datasets

- PBMC 3k (10x): 2,700 blood cells, healthy donor → notebooks 00–04
- Kang 2018: 8 donors, control vs IFN-β → notebook 05

Tools (scverse)

- Scanpy + AnnData for the core analysis
- decoupler, CellTypist, harmonypy, BBKNN, PyDESeq2

AnnData: one object for everything



`adata.layers`
other matrices of the same shape (raw counts)

`adata.obsm`
embeddings per cell (X_pca, X_umap)

`adata.obsp`
cell–cell graphs (neighbors)

`adata.uns`
everything else (colors, parameters, DE results)

`adata.raw`
frozen copy of log-normalized data, all genes

Saved as .h5ad · R equivalents: Seurat, SingleCellExperiment

01 · Quality control: not every barcode is a good cell

Dying / broken cells

Few genes, high % mitochondrial reads. Filter with % MT and min. genes.

Doublets

Two cells in one droplet: too many UMIs and mixed markers. Detect with Scrublet / scDblFinder.

Ambient RNA

Free mRNA in every droplet. Correct with SoupX, CellBender, scAR.

Choosing thresholds

Fixed cut-offs (e.g. 200–2,500 genes, < 5 % MT for PBMCs) or data-driven: > 5 MADs from the median. Tissue matters: neurons and cardiomyocytes have naturally high % MT. Always report what you used.

01–02 · From counts to a low-dimensional space

Raw counts
~30k genes, depth differs per cell

Normalize + log
counts per 10k → log(x+1)

HVGs
keep ~2,000 informative genes

PCA
~30 axes capture the biology

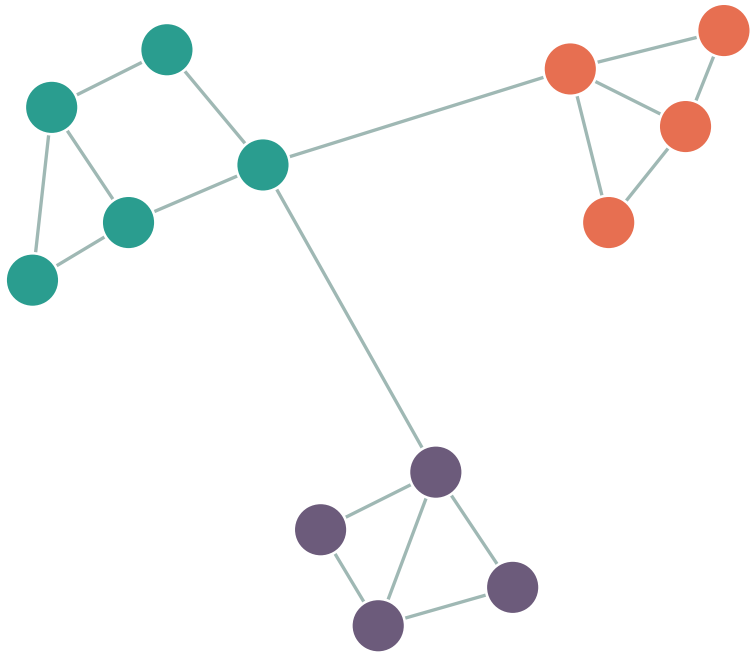
Keep the raw counts!

- adata.layers["counts"] before normalizing
- Pseudobulk DE and scVI need integers
- Alternatives: scran, Pearson residuals, SCTransform

How many PCs?

- Look for the elbow in the variance plot
- A few too many is safer than too few
- PBMC 3k: elbow ~10–15, we use 30

02 · Neighbors, UMAP and Leiden clustering



kNN graph: each cell linked to its 15 most similar cells; Leiden finds densely connected communities

UMAP is a picture, not a measurement

- Local neighborhoods are preserved
- Distances between clusters and cluster sizes are not reliable

Resolution is a choice

- Higher resolution → more, smaller clusters
- Compare several (0.3, 0.8, 1.5) and follow the biology

03 · Three ways to name your clusters

Manual · marker genes

- rank_genes_groups → top genes per cluster
- Compare with known markers (dot plot)
- Transparent, uses your expertise

Marker databases

- PanglaoDB, CellMarker 2.0
- Score every cell (decoupler ULM)
- Fast, but generic lists

References & classifiers

- CellTypist, sc.tl.ingest, scANVI
- R: SingleR, Azimuth
- Reproducible; cannot find new types

Combine them: agreement between independent methods is your evidence; disagreement tells you where to look closer.

03 · Where to find references and record annotations

PanglaoDB

Curated markers, human & mouse · panglaodb.se

CZ CELLxGENE

> 100 M standardized cells; Explorer, Gene Expression, Census API

Single Cell Portal

Broad Institute: hundreds of studies, plots and downloads

Cell Annotation Platform

CAP · celltype.info: record labels with evidence and share them

CellTypist

Pre-trained classifiers: immune, lung, gut, brain…

Cell Ontology

Standard IDs (CL:0000236 = B cell) that make labels reusable

04 · Pick the plot for the question

Plot	Question it answers	Scanpy
UMAP	Where are the cell types? Where is gene X expressed?	sc.pl.umap
Violin / stacked violin	How is a gene distributed in each group?	sc.pl.violin, sc.pl.stacked_violin
Dot plot	Many markers × groups: % expressing and mean level	sc.pl.dotplot
Matrix plot / heatmap	Compact means, or cell-level heterogeneity	sc.pl.matrixplot, sc.pl.heatmap
Proportion bars	How does composition change between samples?	pandas + matplotlib
Volcano plot	Which genes change, how much, how significantly?	matplotlib / decoupler

Tips: colorblind-safe palettes (viridis), labels on the data, vmax="p99" to tame outliers, save as PDF/SVG.

04–05 · Differential expression: which test?

Markers between clusters

- Wilcoxon per cell: rank_genes_groups
- Good for describing and annotating clusters
- p-values are optimistic: genes were used to build the clusters

Conditions (disease vs. control)

- Pseudobulk: sum counts per sample × cell type
- DESeq2 / edgeR with the donor as the replicate
- Needs ≥ 3 biological replicates per condition

Cells from one person are not independent replicates. Treating them as such produces thousands of false positives (Squair et al. 2021).

04 · What do the genes mean?

Over-representation (ORA)

- Top DEGs vs. gene sets (Fisher test)
- MSigDB Hallmark, GO, Reactome
- Simple; depends on a cut-off

Pathway activity

- PROGENy footprints: weighted target genes
- Score per cell or per contrast (ULM)
- Uses direction and strength

Gene signatures

- Any list from a paper
- sc.tl.score_genes per cell
- e.g. interferon response

Beyond today: cell–cell communication (LIANA+, CellChat), trajectories (PAGA, scVelo, CellRank), TF activity (CollecTRI), spatial (Squidpy).

05 · Integration: align samples, keep biology

Method	Works on	Strengths	Languages
Harmony	PCA	Fast, simple, robust default	R · Python
BBKNN	kNN graph	Very fast; graph only (clustering, UMAP)	Python
scVI / scANVI	Raw counts	Complex batches, atlases; label transfer; GPU	Python
Seurat CCA / RPCA	Anchors	Well documented, widely used	R



Batch mixing ↑

Cells of the same type from different samples are neighbors



Bio conservation ↑

Different cell types stay apart (scIB metrics)

R or Python? Both, and you can move between them

Step	Python (scverse)	R (Seurat / Bioconductor)
Object	AnnData (.h5ad)	Seurat (.rds) · SingleCellExperiment
QC · doublets	calculate_qc_metrics · scrublet	PercentageFeatureSet · scDblFinder
Normalize · HVG	normalize_total + log1p · highly_variable_genes	NormalizeData / SCTransform · FindVariableFeatures
Graph · clusters	neighbors · leiden · umap	FindNeighbors · FindClusters · RunUMAP
Markers	rank_genes_groups	FindAllMarkers
Annotation	CellTypist · ingest · scANVI	SingleR · Azimuth
Pseudobulk DE	decoupler + PyDESeq2	muscat · DESeq2 · edgeR

Convert: zellkonverter::readH5AD() · anndata2ri · rpy2. Defaults differ, so results are similar, not identical.



Python

Millions of cells, GPUs, deep learning (scvi-tools), ML pipelines



R

Bioconductor statistics, many specialized methods, polished vignettes

Let's start: Google Colab in 4 steps

1 · Open
github.com/Kur1sutaru/ISMB-LATAM → Open in Colab

2 · Copy
File → Save a copy in Drive

3 · Setup
Run the first code cell (~1 min)

4 · Run
Shift + Enter, cell by cell, in order

Good to know

- 🟢 Core = live · 🔎 Extra = self-study
- Each notebook rebuilds the previous step if needed
- USE_DRIVE = True keeps results in Drive

If something breaks

- Runtime → Restart session and run all
- "Not authored by Google" → Run anyway
- Raise your hand: helpers are around

Keep learning

Single-cell best practices

sc-best-practices.org

Scanpy & scverse tutorials

scanpy.readthedocs.io · scverse.org

OSCA (Bioconductor) · Seurat

bioconductor.org/books/release/OSCA · satijalab.org/seurat

Galaxy Training Network

single-cell track, no code needed

Community

ISCB-LATAM, ISCB Regional Student Groups (RSG-Brasil, RSG-Peru…)

These materials

github.com/Kur1sutaru/ISMB-LATAM

Thank you! ¡Gracias! Obrigada!

Questions, feedback, bugs: open an issue on GitHub

github.com/Kur1sutaru/ISMB-LATAM

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Code: MIT · Slides and text: CC BY 4.0